Homework 1: Semantic Search with Retrieval and Fine-Tuning

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1 Introduction

This report evaluates a dense information retrieval system for semantic search, investigating the mechanics of fine-tuning across two main questions:

- **Question 1(Loss Strategy Comparison):** Can a list-wise strategy using in-batch negatives (Multiple Negatives Ranking) provide a significantly better spatial representation than a pair-wise Contrastive approach relying on mined hard negatives?
- **Question 2 (Architectural Gaps):** Can targeted fine-tuning equalize the performance gap between a generalized multilingual model and specialized English monolingual models on a specific retrieval task?

To this end, we evaluate three neural baselines (DistilBERT, all-MiniLM-L6-v2, and F2LLM-v2-80M) against their fine-tuned counterparts.

2 Methodology

To address the two questions, specific fine-tuning strategies impact over semantic search capabilities have been evaluated. Final models were selected based on the highest Mean Reciprocal Rank (MRR@10) and consistent Hit@K performance on the dev set to ensure generalizability.

To fulfill the Question 1 purpose, two fine-tuning strategies have been chosen. A brief introduction and motivation for both strategies is included below:

- **Contrastive Sentence Pairs (CSP)**: A pair-wise approach minimizing the distance between the query and the correct answer while maximizing the distance from negative samples. We combine explicitly mined hard negatives with random easy negatives to force the learning of fine-grained semantic boundaries. Not only does this approach give more context to the models, but it is also strictly correlated with the other strategy.
- **Multiple Negatives Ranking (MNR)**: A list-wise approach to evaluate an anchor against a positive sentence and a batch of negatives simultaneously. It has been adopted as a fine-tuning strategy because it exploits dataset's design (multiple candidates provided for each query), retrieving a deeper and meaningful context for the model, forcing it to learn fine distinctions between similar sentences. Also, the Multiple Negatives Ranking Loss uses dynamic in-batch negatives, so for each query in the train set, a set of other queries and their correspondent candidates functions automatically as negatives.

While both strategies distance negative samples, MNR relies on a broader, dynamically changing context, whereas CSP relies on static, explicitly mined negatives.

3 Experimental Setup

To mine hard negatives, embedding on query and candidates is handled by 'avsolatorio/GIST-small-Embedding-v0', one of the top performative <100M parameter models for STS tasks, as shown by the Massive Text Embedding Benchmark.

For the Contrastive Sentence Pairs strategy, each instance is formatted as triples (anchor, positive/negative, label). Alongside the positive pair for each query (query, answer), also five negative pairs are generated: three hard negative pairs are given by the top 3 incorrect candidates ranked by GIST cosine similarity, while other two easy negatives are given by two random queries from the dataset.

On the other hand, the Multiple Negatives Ranking strategy requires as input a set of n-tuples including anchor (the query), a positive sentence (correct answer) and the same 3 GIST-ranked incorrect

candidates as for the CSP strategy. Combined with MNR's inherent in-batch negatives, this provides a complex and robust training signal, yielding the possibility of greater performances.

Reproducibility Note. Due to hardware non-determinism across accelerators (e.g., Apple MPS vs. NVIDIA CUDA), re-executing the pipeline may yield marginal Hit@K fluctuations ($\sim$0.01), though relative rankings remain strictly consistent.

4 Results

Three different models were trained using two fine-tuning strategies. To evaluate spatial robustness, both cosine similarity and euclidean distance were computed. Since all vectors were strictly L2-normalized during embedding extraction, both metrics yielded the exact same rankings for Hit@K and MRR@10, so only cosine similarity is reported for the whole evaluation process.

The validation metrics on the dev split (Table 2) guided the hyperparameter selection (detailed in Appendix A.1). Given the marginal performance gaps observed across iterations, the models reached a performance plateau where differences account for a small number of queries. For this reason, the final models have been selected based on the highest MRR@10 and consistent stability between Hit@3 and Hit@5.

The final evaluation on the test set is presented in Table 3. Compared to the zero-shot baselines, a massive improvement is observable across the board.

- **Q1 (Loss Strategy):** The data demonstrates that the Multiple Negatives Ranking strategy consistently yields superior performance compared to the Contrastive Sentence Pairs strategy. By leveraging the full batch as dynamic negative samples instead of relying solely on explicitly mined triplets, MNR dramatically improves the spatial awareness and efficiency of the models. For instance, all-MiniLM jumps from an MRR@10 of 0.7212 (Pairs) to 0.7386 (MNR).

- **Q2 (Architectural Gaps):** As expected, the natively specialized all-MiniLM-L6-v2 achieved the absolute highest scores. However, the most notable finding is the "equalizing" power of the fine-tuning on F2LLM-v2-80M. Despite the architectural disadvantage its MNR fine-tuned version leaped from a baseline Hit@1 of 0.411 to 0.5905. It achieved an MRR@10 of 0.7205, with a very small 0.018 gap from all-MiniLM's result. This proves that targeted fine-tuning can forcefully re-align a multilingual semantic space to excel in a monolingual task.

- **Qualitative Error Analysis:** To understand the residual 0.018 gap between all-MiniLM and F2LLM, the discordant queries have been inspected manually. It has been observed that F2LLM struggled primarily with highly specific English terminology in specialized domains. For example, given the query *"where does absorption take place in the small intestine"*, F2LLM ranked the correct answer 20th, whereas all-MiniLM retrieved it as the 6th most likely candidate. This highlights a gap in specialized monolingual scientific vocabulary, which could motivate the gap. Conversely, F2LLM outperformed all-MiniLM on queries rich in global entities or geographical facts (e.g., *"where does british columbia get its oil from"*), successfully retrieving the answer while all-MiniLM relegated it to the 20th position. This suggests that broad, common-knowledge entities are highly robust in F2LLM's multilingual pretraining space, partially compensating for its monolingual vocabulary limits. More examples can be seen in detail in Appendix A.3.

- **Limitations:** The dataset's restricted scale and domain specificity may limit real-world generalizability. Furthermore, metrics like Hit@K are binary approximations: they capture exact positional precision but fail to reward semantically acceptable (yet unlabelled) candidate chunks, potentially underestimating the models' true semantic capabilities.

Conclusion. Ultimately, while selecting an architecture natively adapted to the task (e.g., all-MiniLM) yields the highest absolute performance, our findings highlight that sophisticated fine-tuning acts as a powerful equalizer. Specifically, leveraging a list-wise negative sampling strategy (MNR) drastically bridges the semantic gap between general-purpose multilingual models and highly specialized monolingual ones.

A Appendix

A.1 Models' results

Baseline model	Hit@1	Hit@3	Hit@5
distilBert-base-uncase	0.1765	0.4435	0.6045
all-MiniLM-L6-v2	0.4480	0.7170	0.6045
F2LLM-v2-80M	0.411	0.696	0.7955

Table 1: Results of baseline models.

Trained model	Hit@1	Hit@3	Hit@5	MRR@10
distilBert-1-pairs	0.5913	0.7925	0.8625	0.7052
distilBert-1-mnr	0.6050	0.8187	0.8962	0.7245
distilBert-2-pairs	0.5950	0.7863	0.8638	0.7061
distilBert-2-mnr	0.6012	0.8125	0.8912	0.7202
distilBert-3-pairs	0.5800	0.7775	0.8550	0.6959
distilBert-3-mnr	0.5988	0.8200	0.8925	0.7195
all-Mini-1-pairs	0.6075	0.8037	0.8750	0.7229
all-Mini-1-mnr	0.6238	0.8375	0.9137	0.7431
all-Mini-2-pairs	0.6100	0.8113	0.8862	0.7266
all-Mini-2-mnr	0.6312	0.8337	0.9125	0.7478
all-Mini-3-pairs	0.6138	0.8113	0.8838	0.7293
all-Mini-3-mnr	0.6200	0.8450	0.9200	0.7435
F2LLM-1-pairs	0.5713	0.7850	0.8575	0.6948
F2LLM-1-mnr	0.6062	0.8125	0.8938	0.7270
F2LLM-2-pairs	0.5750	0.7850	0.8575	0.6972
F2LLM-2-mnr	0.5938	0.8063	0.8875	0.7169
F2LLM-3-pairs	0.5713	0.7837	0.8562	0.6949
F2LLM-3-mnr	0.5925	0.8113	0.8912	0.7200

Table 2: Results on dev split of different trained models.

Training parameters of the different models:

- **model 1**: learning rate=4e-5, batch size=8, gradient accumulation steps=2 weight decay=0 epochs=2, warmup steps=0.1
- **model 2**: learning rate=2e-5, batch size=8, gradient accumulation steps=2, weight decay=0, epochs=3, warmup steps=0.1
- **model 3**: learning rate=3e-5, batch size=8, gradient accumulation steps=2, weight decay=0.1, epochs=2, warmup steps=0.1

Selected model	Hit@1	Hit@3	Hit@5	MRR@10
distilBert-pairs	0.5900	0.7980	0.8800	0.7111
distilBert-mnr	0.5940	0.8255	0.9065	0.7239
all-MiniLM-pairs	0.6010	0.8185	0.8910	0.7212
all-MiniLM-mnr	0.6105	0.8450	0.9140	0.7386
F2LLM-pairs	0.5615	0.8005	0.8745	0.6937
F2LLM-mnr	0.5905	0.8170	0.8970	0.7205

Table 3: Results on the test set of the selected models.

A.2 Models' choices detailed explanation

For DistilBERT, model 2 (CSP) and model 1 (MNR) were selected for their more robust performances. Regarding all-MiniLM-L6-v2, the models' performances are largely comparable; by evaluating stability across Hit@5 and Hit@3, and selecting the highest score on MRR@10, we retained model 3 (CSP) and model 2 (MNR) for their superior generalization. Finally, for F2LLM-v2-80M, the chosen models are model 2 for CSP, which outperformed other variants, and model 1 for MNR, which exhibited superior performance. Given the marginal performance gaps observed, the models appear to have reached a performance plateau where differences account for only a small fraction of queries. Although we selected the final models based on the highest values over the chosen metrics to ensure a robust ranking, we acknowledge that at this scale, variations might be subject to statistical noise, so some models not selected for test set evaluation could still perform well, and in some cases even better than the selected ones.

A.3 MNR strategy comparison: AllMini-LM vs F2LLM

Query ID	Query Text	all-MiniLM Rank	F2LLM Rank
1VssDNBiFGhn	*where does absorption take place in the small intestine*	6	20
fhNg3t2nEqai	*where does british columbia get its oil from*	20	1
cEEFKGx1GEso	*when did twenty one pilots start the hiatus*	3	23
JOxYh3zChU9z	*when did the only way is essex first start*	21	2
PJAD39QctFfw	*who had the first dunk in nba history*	6	20
FVr9Qh8yXv6U	*who was the owner of the union pacific railroad*	18	5

Table 4: Examples of discordant retrievals between all-MiniLM and F2LLM.